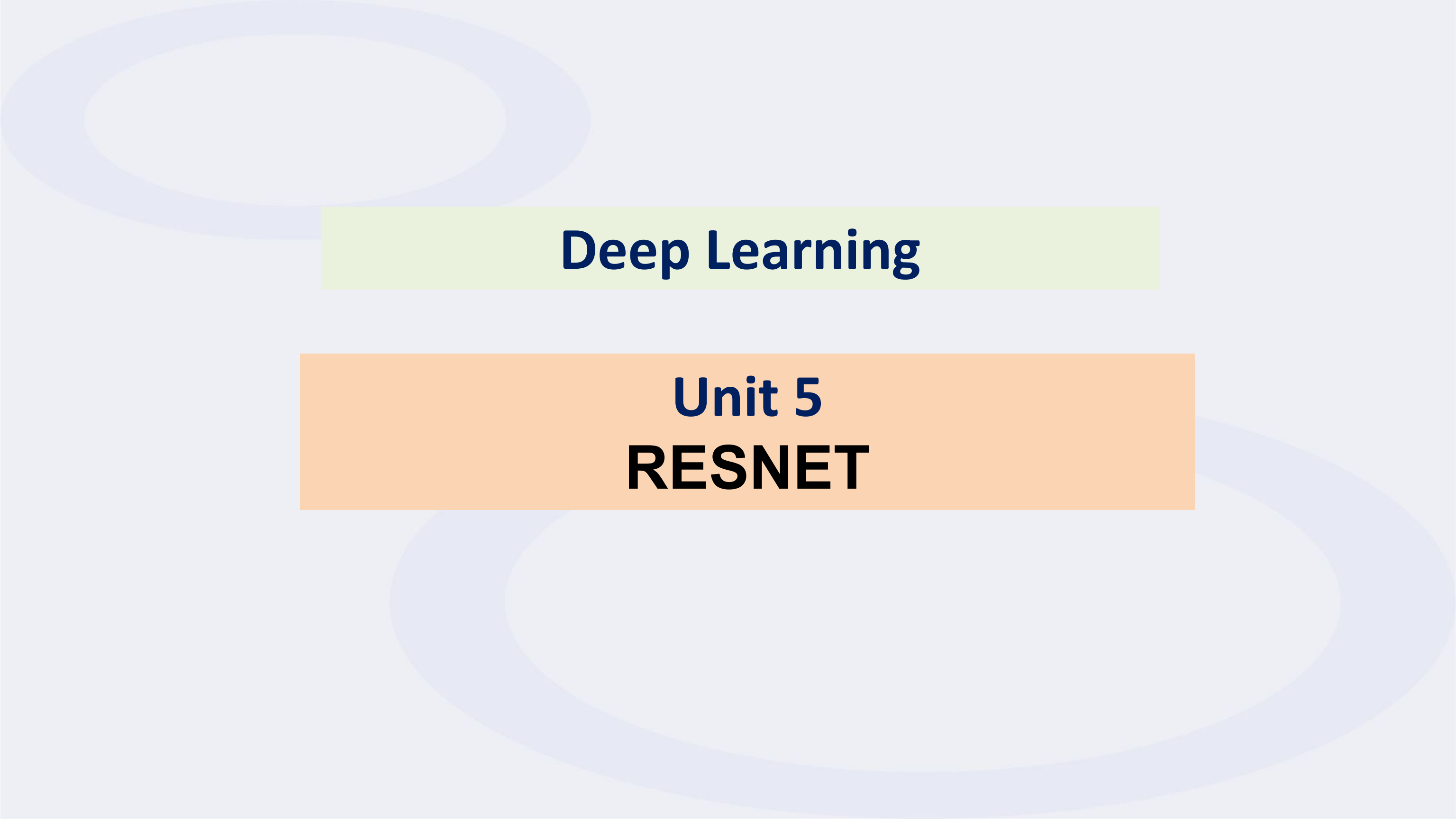
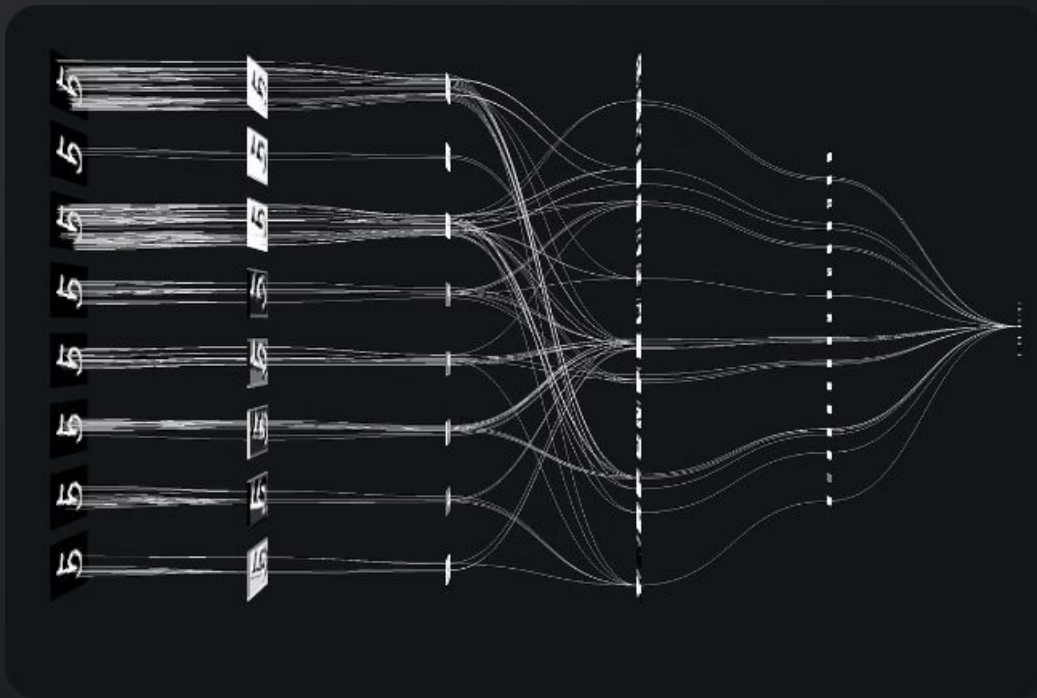




Deep Learning

Unit 5

RESNET



Introduction to ResNet

This presentation delves into the intricacies of ResNet, a groundbreaking deep learning architecture that has revolutionized the field of computer vision. ResNet's unique design addresses the challenges of training deep neural networks, allowing for the creation of exceptionally powerful and accurate models.



What is ResNet?

ResNet, short for Residual Neural Network, is a type of artificial neural network that excels at image recognition and related tasks. It employs residual connections, a novel architectural feature, to effectively learn and represent complex patterns within data.

Deep Learning

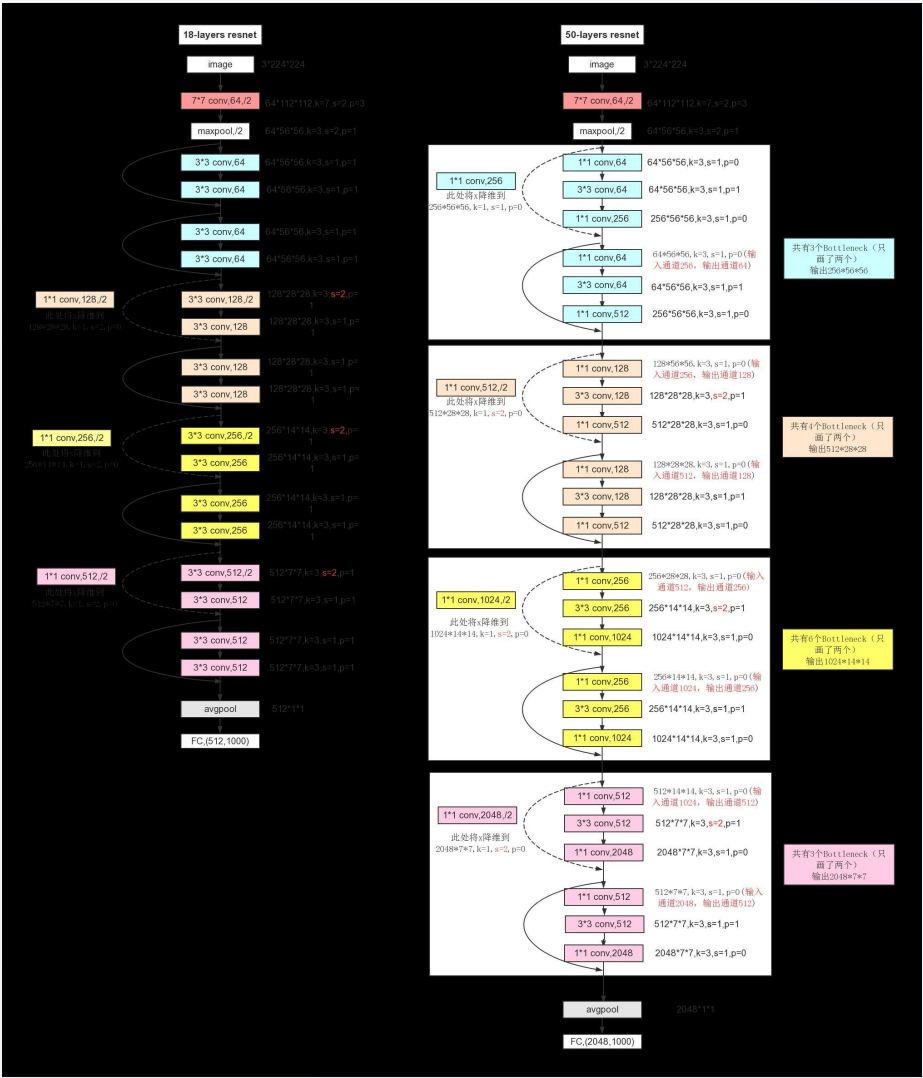
ResNet is a deep learning architecture, meaning it utilizes multiple layers to extract hierarchical features from data.

Image Recognition

ResNet has proven to be highly effective in image recognition tasks, achieving state-of-the-art performance.

Residual

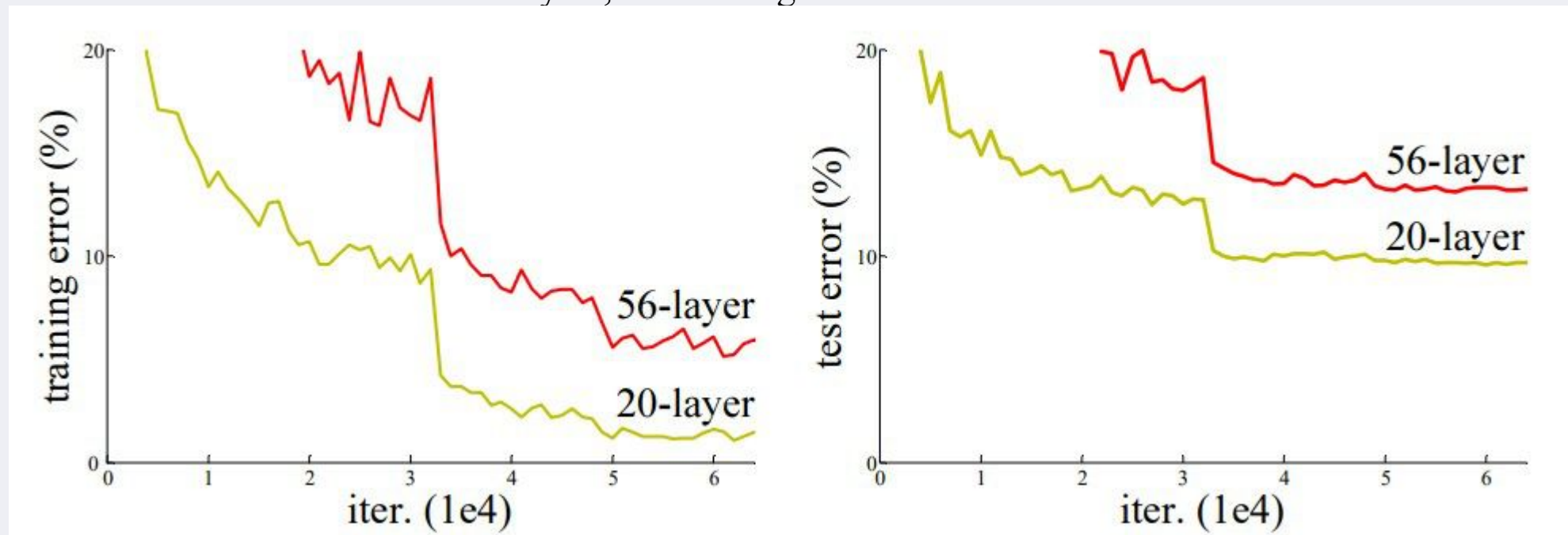
Connections is the introduction of residual connections, which facilitate the flow of information across layers.



Motivation behind

ResNet

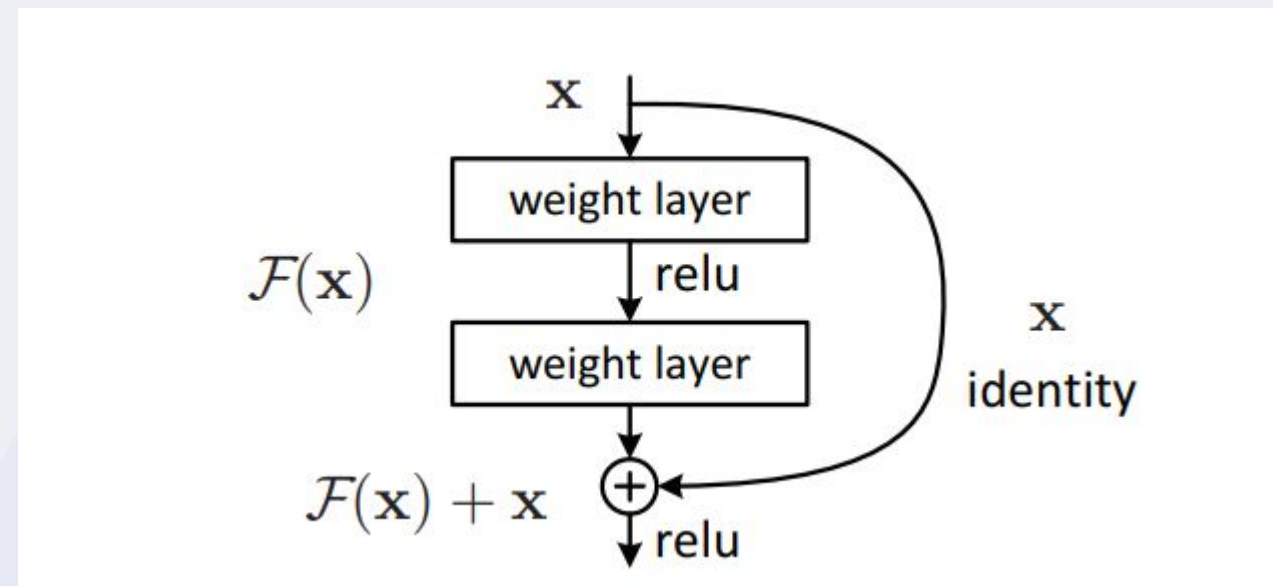
After the first CNN-based architecture (AlexNet) that won the ImageNet 2012 competition, Every subsequent winning architecture uses more layers in a deep neural network to reduce the error rate. This works for less number of layers, but when we increase the number of layers, there is a common problem in deep learning associated with that called the Vanishing/Exploding gradient. This causes the gradient to become 0 or too large. Thus when we increase number of layers, the training and test error rate also increases.



In the above plot, we can observe that a 56-layer CNN gives more error rate on both training and testing dataset than a 20-layer CNN architecture. After analyzing more on error rate the authors were able to reach conclusion that it is caused by vanishing/exploding gradient. ResNet, which was proposed in 2015 by researchers at Microsoft Research introduced a new architecture called Residual Network.

Residual Network: In order to solve the problem of the vanishing/exploding gradient, this architecture introduced the concept called Residual Blocks. In this network, we use a technique called *skip connections*. The skip connection connects activations of a layer to further layers by skipping some layers in between. This forms a residual block. Resnets are made by stacking these residual blocks together. The approach behind this network is instead of layers learning the underlying mapping, we allow the network to fit the residual mapping. So, instead of say $H(x)$, initial mapping, let the network fit,

$$F(x) := H(x) - x \text{ which gives } H(x) := F(x) + x.$$



Motivation behind

ResNet

As neural networks become deeper, they encounter the vanishing gradient problem, where gradients diminish during backpropagation, hindering effective learning. ResNet was developed to address this challenge and enable the training of deeper networks.

Vanishing Gradients

In deep networks, gradients can become vanishingly small, preventing the network from learning effectively.

Residual Connections

ResNet introduces residual connections to bypass layers and allow gradients to flow more easily, mitigating the vanishing gradient problem.

Deeper Networks

With ResNet, it becomes feasible to train significantly deeper networks, leading to improved performance.

Architecture of

ResNet

The core of ResNet lies in its residual blocks, which consist of stacked convolutional layers. These blocks are connected through skip connections, allowing information to flow directly from earlier layers to later layers, bypassing intermediate layers.

1

Convolutional Layers

ResNet utilizes convolutional layers to extract features from input data, such as images.

2

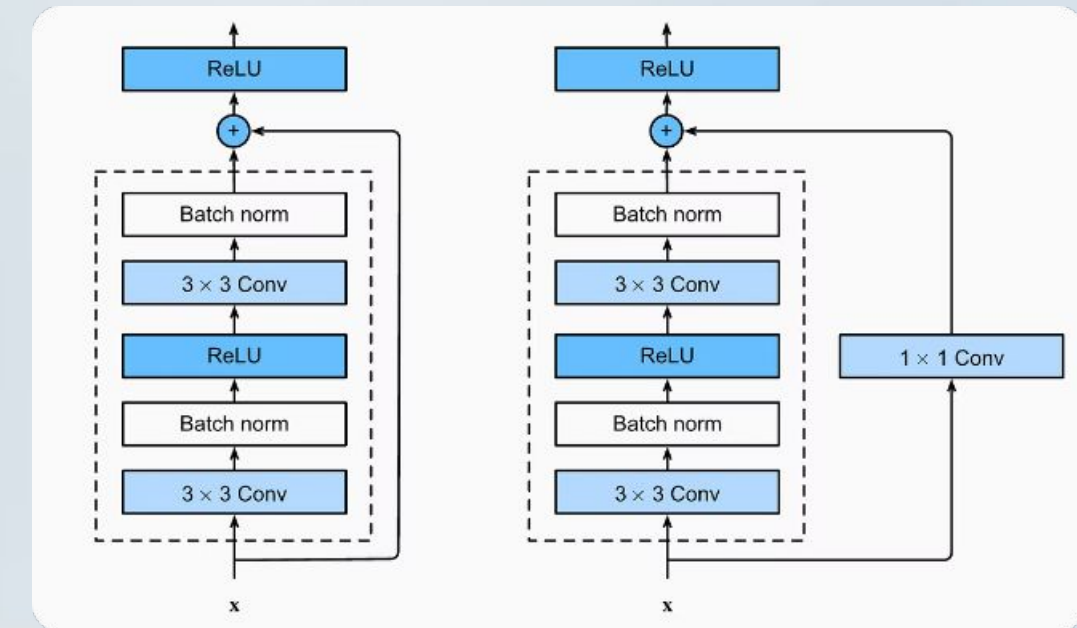
Residual Blocks

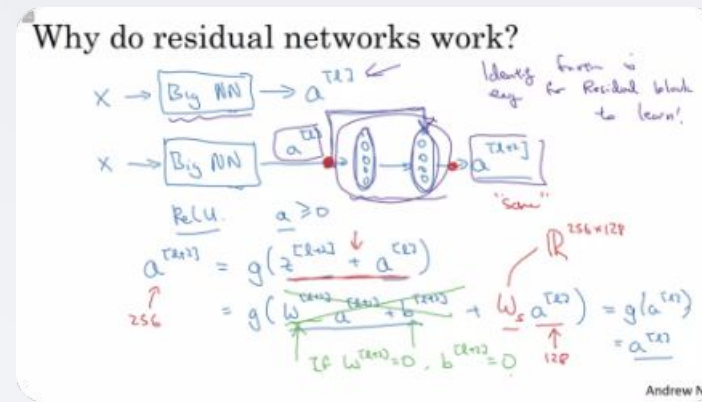
Residual blocks are stacked convolutional layers that form the building blocks of the ResNet architecture.

3

Skip Connections

Skip connections enable information to bypass layers and flow directly to later layers, preserving gradient information.





Residual Connections

Residual connections are the key innovation in ResNet. They allow the network to learn an identity mapping, which essentially enables the direct transfer of information from earlier layers to later layers. This allows gradients to flow more easily, preventing vanishing gradients and enabling the training of deeper networks.

1

Input

The input data is fed into the ResNet architecture.

2

Residual Block

The input is passed through a residual block, containing convolutional layers.

3

Skip Connection

The input is also directly passed through a skip connection to the output.

4

Output

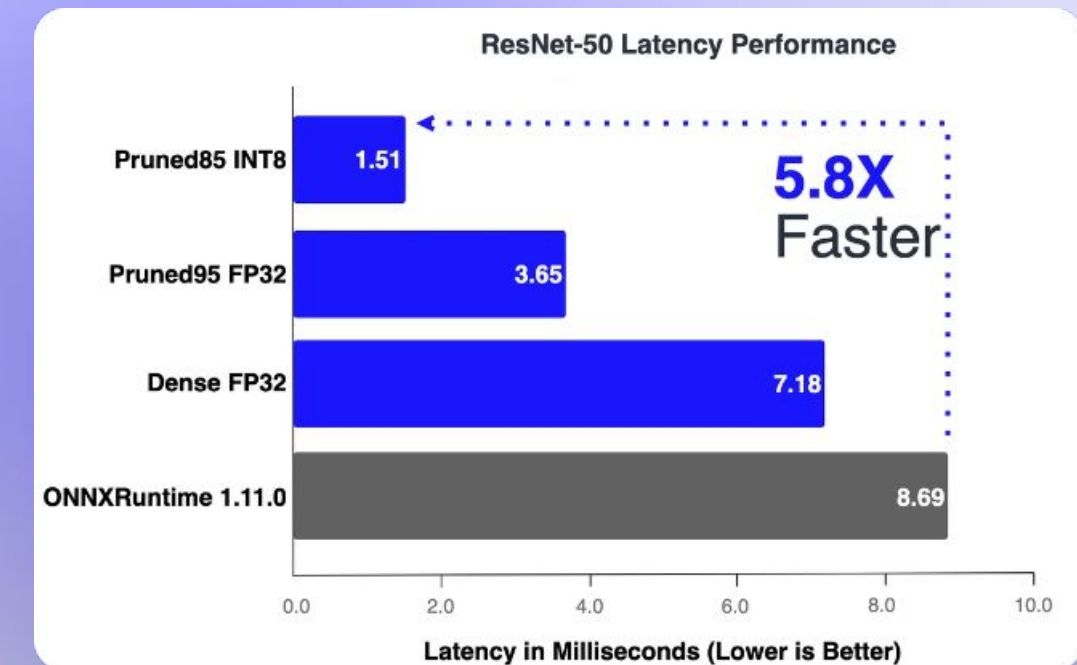
The output is the sum of the outputs from the residual block and the skip connection.

Advantages of

ResNet offers several advantages over traditional deep neural networks, making it a preferred choice for image recognition tasks and other applications.

- 1 Improved Performance**
ResNet significantly outperforms traditional deep networks in image recognition tasks, achieving state-of-the-art accuracy.
- 3 Reduced Overfitting**
ResNet's architecture helps mitigate overfitting, leading to models that generalize better to unseen data.

- 2 Deeper Networks**
ResNet enables the training of much deeper networks, allowing for the extraction of more complex features from data.
- 4 Faster Training**
The use of residual connections in ResNet can accelerate the training process, making it more efficient.



Deep Learning for Computer Vision: Models & Real World Applications



Applications of ResNet

ResNet's versatility has led to its widespread adoption in numerous applications across diverse fields, particularly in computer vision. Its exceptional performance has propelled advances in various

domains.
Image

Recognition

Object Detection

Image

Segmentation

Medical Imaging

Self-Driving Cars

Robotics



Deep Learning Technology;
The Future Is Going To Get A Lot Easier

Conclusion and Future

Directions

ResNet has profoundly impacted the field of deep learning, establishing a benchmark for image recognition and influencing the design of subsequent architectures. Ongoing research continues to explore variations and extensions of ResNet, pushing the boundaries of deep learning capabilities.



Further Research

Exploring novel variations of ResNet to enhance performance and address specific challenges.



Broader

Applications Exploring ResNet's capabilities to address a wider range of tasks, such as natural language processing and time series analysis.



Performance

Optimization Exploring techniques to further optimize ResNet's training and inference efficiency.

THANK YOU